

G9-COMPACTNESS-AUTO-AUDIT-v1: Does $T^* < \infty \Rightarrow$ suitable-weak compactness of u_n ?

ZFL-CX

September 3, 2026

$G7 = FROZEN$, $G8 = CLOSED/BLOCKED$, $G9-CHECK12 = BLOCKED$ at Step2

$$Q : T^* < \infty \stackrel{?}{\Rightarrow} u_n = r_n u(x_n + r_n y, t_n + r_n^2 s) \rightarrow \bar{u}$$

No Type I, no $L_t^\infty L^3$, no $\bar{\alpha} > 0$, no profile hypothesis, no artificial r_n rate.

1 Scaling of Leray-Hopf bounds - supercriticality

For $u_n = r_n u(x_n + r_n y, t_n + r_n^2 s)$:

$$\int_{B_R} |u_n(y, s)|^2 dy = r_n^{-1} \int_{B_{r_n R}(x_n)} |u(x, t_n + r_n^2 s)|^2 dx.$$

Global bound $\|u\|_{L_t^\infty L_x^2}^2 \leq E_0$ gives only:

$$\|u_n(s)\|_{L^2(B_R)}^2 \leq r_n^{-1} E_0 \rightarrow +\infty.$$

Similarly $\int_{-S}^0 \|\nabla u_n\|_{L^2(B_R)}^2 ds = r_n^{-1} \int \|\nabla u\|_{L^2(B_{r_n R})}^2 dt$. Thus Leray-Hopf global energy is supercritical: does NOT give uniform local bound after blow-up scaling.

Consequence: suitable-weak existence $\neq$ compactness of blow-up sequence.

Local energy inequality for suitable u :

$$\|u(t)\|_{L^2(B_r)}^2 + \int \|\nabla u\|^2 \leq Cr \cdot (\text{terms}) + \text{pressure}.$$

Scaling: RHS $\sim r$, LHS after scaling $\sim r^{-1}$ factor. No automatic $C(R, S)$.

2 Gate 1: Existence of limit

Question: Does there exist R, S s.t. $\|u_n\|_{L_t^\infty L^2(B_R)} + \|\nabla u_n\|_{L^2(B_R \times (-S, 0))} + \|p_n\|_{L^{3/2}} \leq C(R, S)$ uniformly?

With only $T^* < \infty + C_E^{weak}$ (suitable), answer: NO in general. Known blow-up constructions (Escauriaza-Seregin-Sverak) require Type I $r_n \sim \sqrt{T^* - t_n}$ to get uniform local energy bound. Without it, $r_n^{-1} E_0$ blows up.

Possible rescue: Use ε -regularity + smallness of scaled energy? At singular point, scaled energy $\limsup r^{-1} \int_{B_r} |u|^2 \geq \varepsilon_{CKN}$. This gives LOWER bound, not upper.

Thus Gate1 requires additional quantitative small-scale control beyond $\mathcal{C}_{PROVEN}$. With only $\{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\}$, Gate1 = BLOCKED.

Exception: If r_n chosen to balance local L^2 mass, i.e., $r_n^{-1} \int_{B_{r_n}} |u(t_n)|^2 \sim 1$, then uniform $L^2(B_1)$ holds by construction, but then r_n is no longer linked to C_F^{conc} scale $\|u\|_{L^3(B_{r_n})} \geq \varepsilon_0$ nor to N_c^{-1} . So scale coincidence not proved.

3 Gate 2: Preservation of concentration

Assume for contradiction Gate1 passed with $u_{n_k} \rightharpoonup \bar{u}$ weakly in L_{loc}^2 or weak-* suitable.

We have:

$$\|u_n(0)\|_{L^3(B_1)} = \|u(t_n)\|_{L^3(B_{r_n})} \geq \varepsilon_0.$$

Weak convergence $u_n \rightharpoonup \bar{u}$ in L^2 does NOT preserve $L^3(B_1)$ lower bound. L^3 norm is not weakly continuous; concentration can vanish via oscillation or escape to frequency $N_c \rightarrow \infty$.

C_I^{freq} says frequency diverges, which actually threatens preservation: energy may move to $|\xi| \gg r_n^{-1}$, invisible to weak L^2 limit.

Thus even IF Gate1 held, we need strong convergence $u_n \rightarrow \bar{u}$ in L_{loc}^3 or at least $L^3(B_1)$ lower semicontinuity with concentration retained. This requires uniform L^3 bound + compactness (Rellich), which we lack.

Hence:

$$\text{weak } u_n \rightharpoonup \bar{u} \not\Rightarrow \|\bar{u}\|_{L^3(B_1)} \geq \varepsilon_0$$

$$\text{concentration} \not\Rightarrow \text{preserved under weak limit}$$

4 ZFL verdict

Gate1 (existence of suitable-weak limit with uniform local bounds) = BLOCKED with $\mathcal{C}_{PROVEN}$

Gate2 (preservation of ε_0) = BLOCKED even if Gate1 rescued, without L_{loc}^3 strong convergence

Therefore:

$$\text{G9 direct } T^* < \infty \Rightarrow \bar{u} \text{ ancient non-trivial} = \text{BLOCKED}$$

This is excellent ZFL result: we did not spend time fabricating ancient profile that current hypotheses cannot build.

Next only if new critical-scale bound producing compactness is discovered, or different mechanism giving weak compactness + rigidity without full critical bound. No hypothesis smuggling.

G7 FROZEN $\rightarrow$ G8 CLOSED $\rightarrow$ G9-CHECK12 BLOCKED $\rightarrow$ G9-AUTO-COMPACTNESS BLOCKED

$$\text{STOP} = \text{PROGRESO}$$